Case Study: Operation InVersion at LinkedIn (2011)

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Introduction

The Case Study: Operation InVersion at LinkedIn (2011) was focused on a time when technical debt was not repaid and it had significant consequences. This case study was written to demonstrate the importance of addressing technical debt as a part of everyday development. There were major issues that arose which led to a halt in developing new features. The case study explains that, due to problematic deployments, “they stopped all feature development for two months in order to overhaul their computing environments, deployments, and architecture” (Kim et al., 2021). Two months is not a short period in the world of development.

Case Summary

LinkedIn’s issues arise from simply outgrowing the infrastructure it was built on. Using Java, they cooked up their Leo application to create LinkedIn. LinkedIn was founded in 2003, and its membership quickly grew from around 2,700 members after its first week, exponentially to around 350 million members by November 2015 (Clemm, 2015). Traffic on the site would generate millions of queries and Leo could not keep up with it all. There was an attempt to scale the processing power by adding memory and CPUs, but that was a quick fix that could not be scaled up continuously. It was quickly approaching its limits, and Leo would often go down, making it a challenge to not only recover, but to also develop new features for the site.

Additionally, when new releases came, they would break the site routinely. There is a quote from Ashlee Vance of Bloomberg in 2013: “when LinkedIn would try to add a bunch of new things at once, the site would crumble into a broken mess, requiring engineers to work long hours into the night and fix the problems (Vance, 2013). This was not a sustainable system for the site or the developers involved, something needed to change and change quickly. The way the

infrastructure of the site was made did not allow it to scale gracefully while developing new features and fixing bugs.

Project Inversion

An idea began to creep up called Project Inversion. Kevin Scott, who at the time was VP of Engineering at LinkedIn, along with other top engineers, proposed a stop to all development of new features to focus elsewhere. The plan was to have "every able body focus on rebuilding the company's infrastructure from scratch" (Vance, 2013). This ended up taking about two months to complete and was a direct payment towards their technical debt that had accumulated so far.

This also was time when the work culture shifted. There was a focus on a more sustainable mindset with development. During Project Inversion, they also worked on creating new tools and software to help streamline development. The new tools and automated systems helped identify bugs and compatibility issues before deployment. This made releases safer and allowed developers to instead focus on creating new features. As a result, LinkedIn can have "major upgrades to the site three times a day" (Vance, 2013). This was a miraculous improvement from having troubling deployments.

What Was Learned

This case study was a prime example of how important it is to manage technical debt. LinkedIn started as a much smaller service than it is currently. It was said there were around 2,700 members after the first week, with around a million a year later. This number exploded to 350 million in 2015 (Clemm, 2015). LinkedIn was not initially built with this much user activity in mind. Leo initially served the needs of LinkedIn, but as the company grew, the monolithic application became increasingly difficult to maintain, troubleshoot, and deploy. What worked for

that smaller platform was no longer suitable for the scale LinkedIn had reached. Leo became a hinderance to development and the team needed to shift away from it.

One attempt to fix their issues was vertical scaling, which involved adding more processing power through additional CPUs and memory. This was a quick fix that bought time, but it should have been a time to have a more purposeful development to start taking place. Adding processing power is not something that can be done indefinitely. An opportunity arose here where optimizations should have been delved into. Think about it like this, if the kitchen sink is full of dirty dishes, would you just buy new plates to use instead of washing the ones in the sink? That is basically how the LinkedIn team fixed their issues when adding more CPUs and memory.

Instead of continuing to add more processing power, LinkedIn needed to address the underlying problems with its architecture and deployment processes. This is why technical debt should be managed as a part of everyday development instead of being left until it becomes a major obstacle.

Adding new features is important, but developers need time to also make improvements to existing systems, fix recurring problems, and streamline development. Project Inversion demonstrated that investing in these areas can improve reliability, developer productivity, and the company's ability to grow. While LinkedIn did have to stop feature development for two months to make these changes, addressing the problems directly allowed the team to perform major upgrades three times a day. Overall, teams should learn from this about how crucial it is to continuously work to make improvements before their technical debt requires its own Project Inversion.

References

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